

I N D E X

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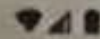
Completed
[Signature]

*Wherever you see "Cover", it means both the front and back.

ANSWER 1
1. AIM FOR TR
2. TRUST YOL
3. FIND YOUR
4. FIRE MEETS
5. NET LIGHTS
6. STARS TRA
7. ON THIS CC
8. KEEP THE E

GKC1125601 C

7:59



LAB 1

Build First Android App

Enter first text

Hello

Enter second text

Hello

Check Match

Correct

Ex: 1

String Comparison

AIM:-

To check whether the first text and second text are the same by considering uppercase and lowercase characters.

ALGORITHM:-

- 1) Start the programme
- 2) Input the first string
- 3) Input second string
- 4) Compare both string characters by character.
- 5) If both strings are identical "Correct", otherwise "Incorrect".

~~By~~
RESULT:-

Thus, the programme executed successfully on string comparison.

8:26

EXPERIMENT 2

94%

LAYOUT



Line

Circle

Show All

Clear All

Ex: 2

GUI Layout with Line Button and Circle Button Visualization.

AIM:-

To check whether and design a graphical user interface with button to draw a line and a circle to visualize them using Show and Clear buttons.

ALGORITHM:-

- 1) Start the program and create GUI
- 2) Add buttons for Line, Circle, Show, Clear
- 3) Line button pressed, generate line shape
- 4) Circle button pressed, generate circle shape.
- 5) Use "Show All" to display and "Clean All" to remove shapes.

~~By~~
RESULT:-

Thus the programme executed successfully on GUI layout

LAB 3

App Navigation

Go to Second Page

Click the button above to navigate to another page

Ex: 3

App Navigation Between Pages.

AIM:-

To implement navigation between the main page and second page using buttons.

ALGORITHM:-

- 1) Start application create main page
- 2) Add a button "Go to Second Page" on main page.
- 3) Button click navigate to second page
- 4) On second page add "Main Page" button
- 5) Button click navigate to Main Page.

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RESULT:

Thus the programme executed successfully on Navigation App

LAB 4

Activity and Fragment Lifecycles

Simple Timer

00:00:00.00

START

PAUSE

STOP

Activity Lifecycle Events

```
[1773328228547] onCreate  
[1773328228706] onStart  
[1773328228715] onResume
```


Ex : 4

Activity and Fragment Lifecycle with Simple Timer.

AIM:-

To demonstrate Activity and Fragment and implement a simple timer with Start, Pause and Stop buttons.

ALGORITHM:-

- 1) Start application and create an activity with a fragment and a timer display (00:00:00:00)
- 2) Display lifecycle on Create(), on Resume(), OnStart when activity starts
- 3) Add "Start" button to begin timer counting
- 4) Add "Pause" button to temporarily stop the timer.
- 5) Add "Stop" button to reset timer.

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RESULT:-

Thus the programme executed successfully on lifecycle.

LAB 5

App Architecture (UI Layer)

Button 1

Button 2

Card Example

This card shows padding
and margins

Rounded Box
with padding



Ex: 5

App Architecture - UI
Layer with Buttons and
Card.

AIM:-

To design a simple UI layer using buttons and a card layout with a rounded box, padding and different colours.

ALGORITHM:-

- 1) Start application and create UI Layout
- 2) Add Button 1, Button 2 to interface
- 3) Create a card component with rounded box and apply padding
- 4) Apply different background colours such as red, green and blue to UI
- 5) Display UI layer with buttons and card.

RESULT:-

Thus the programme executed successfully on UI layout.

LAB 6

App Architecture - Persistence

SharedPreferences

Saved Username:

John Doe

Internal Storage

Saved Note:

This is a sample note stored in internal storage.

Room Database (Tasks)

- Complete Android Lab
- Study Jetpack Compose
- Prepare Lab Screenshot

Persistence Methods Used

- SharedPreferences – Key-value storage
- Internal Storage – File based storage
- Room Database – SQLite structured storage

Ex. 16

App Architecture Persistence Methods.

AIM:-

To demonstrate data persistence in applications using Shared Preferences, Internal Storage and Room database.

ALGORITHM:-

- 1) Start the application and create basic app
- 2) Use Shared Preferences to store small key-value data
- 3) Use Internal Storage to save and retrieve data files inside the device
- 4) Implement Room Database to store and manage tasks in structured databases.

~~bo~~
RESULT:-

Thus the programme executed successfully on persistence methods.

9:02



LAB 7

Advanced RecyclerView Use Cases

GETTING STARTED



Basic RecyclerView

Setting up a simple list with a single view type



View Binding

Using ViewBinding for cleaner view access in Viewholders

ADVANCED ARCHITECTURES



Multiple View Types

Combining Headers and Content items in one list



DiffUtil & ListAdapter

Optimizing list updates with efficient background diffing

INTERACTIONS



ItemTouchHelper

Implementing Swipe-to-Delete and Drag-and-Drop functionality



Click Listeners

Handling complex item interactions and navigation

PERFORMANCE

Ex : 7

Advance Recycle view use cases

AIM:-

To implement Advance RecyclerView features including multiple view types, interactions and performance optimization.

ALGORITHM:-

- 1) Start application and create a basic RecyclerView and View Binding
- 2) Implement multiple view types and manage data using DiffUtil and List Adapter.
- 3) Add user interaction : ItemTouchHelper and clickListeners.
- 4) Improve performance by optimizing data updates and view recycling.
- 5) Display list items efficiently using advanced RecyclerView architecture.

RESULT:-

Thus the programme executed successfully on View Recycler.

9:03



LAB 8

Connect to the internet



Connected

Understanding Connectivity

This app uses `ConnectivityManager` and `NetworkCallback` to listen for real-time network changes. When the device gains or loses internet access, the UI updates automatically.

Ex : 8

Internet Connectivity Detection Using Connectivity Manager

AIM:-

To detect internet connectivity and update UI when network status change.

ALGORITHM:-

- 1) Start application and check internet connection
- 2) Use Connectivity Manager to monitor the network
- 3) Implement Network Callback to listen for real time - network changes.
- 4) Detect gains or losses internet access
- 5) Update UI automatically based on connectivity status.

RESULT:

Thus the programme executed successfully on Connectivity Manager.

Ex: 9 Repository Pattern and Work Manager Implementation

AIM:-

To implement Repository Pattern and Work Perform to perform background synchronization of data.

ALGORITHM:-

- 1) Start application and implement Repository Pattern to manage data operations.
- 2) Use WorkManager to schedule background tasks.
- 3) Create a SyncWorker that runs in background to add or sync new goals.
- 4) Execute the worker at specific time and data.
- 5) Update UI automatically using Flows when new data syncd.

RESULT:-

This the programme executed successfully on Synchronization.

LAB 9

Repository Pattern and Work Manager

How it works:

1. Repository manages data from Room DB
2. Work Manager runs SyncWorker in background
3. UI updates automatically via Flows.

New Goal Sync

This goal was added in the background by Work Manager.

Synchronised at: 14:55:00, 12 Mar 2024

New Goal Sync

This goal was added in the background by Work Manager.

Synchronised at: 14:55:00, 12 Mar 2024

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Synchronised at: 14:55:00, 12 Mar 2024

New Goal Sync

This goal was added in the background by

Synchronised at: 14:55:00, 12 Mar 2024

+ Sync Now

Ex: 10

App UI Design Principles.

AIM:

To understand and apply basic UI design principles in application.

ALGORITHM:-

- 1) Start application and design a UI
- 2) Apply color theory to choose suitable colors
- 3) Use proper typography for readable.
- 4) Maintain proper layout and spacing for better UI
- 5) Add UI components buttons, cards and text elements.

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RESULT:-

~~This~~ the programme executed successfully on Design Principles.

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LAB 10

App UI Design

Learn & Understand UI Principles



Color Theory

Harmony, Contrast & Palette



Typography

Fonts, Sizes & Readability



Layout & Spacing

Grids, Padding & Alignment



Components

Interactive UI Elements